

Mapping the Galactic Dust Map using RR Lyrae Stars

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Used Gaia DR3 data [1] to analyze RR Lyrae variable stars and map interstellar dust in the Milky Way. This was possible since RR Lyrae stars serve as standard candles through their well-defined period–luminosity and period–color relations. By validating Gaia’s reported pulsation periods using Lomb–Scargle periodograms and modeled light curves with Fourier series to estimate mean magnitudes. Cross-validation determined that four Fourier terms ($K = 5$) was the best to describe the data. Using a sample of 400 RR Lyrae stars with reliable parallaxes and $b - i < 30^\circ$, I then derived a G-band period–luminosity relation of $M_G = 0.14 - 2.95 \log P$, consistent in slope with literature after accounting for bandpass and extinction differences. Bayesian MCMC (both Metropolis–Hastings and NUTS) confirmed the model’s robustness and scatter. By extending this to a period–color relation to estimate intrinsic colors and derive reddening $E(\text{GBP} - \text{GRP})$, constructing a Galactic dust map. The map reproduces the general morphology of the Schlegel–Finkbeiner–Davis (SFD) map but shows lower amplitudes, reflecting its distance-limited nature. The results demonstrate the utility of RR Lyrae stars as tracers of 3D dust distribution and the power of Gaia data for Galactic structure studies.

I. INTRODUCTION

RR Lyrae can be identified as stars that are low mass, old and whose surfaces expand and contracts regularly with periods shorter than a day. With this pulsation this gives a variation of brightness occurring in two main flavors. Those being c-type RR Lyrae (RRc) and ab-type (RRab). RRc can be described by their close sinusoidal light curves, that repeat in time with periodicities that vary between 0.2 and 0.42 day. Also their amplitude of the light variation can go up to about half a magnitude in the visual band. With RRab stars they are described by having asymmetrical sawtooth light curves and their periods are around 0.42 and 1 day. The amplitudes in the Gaia B band are found to be above 0.1 to more than one magnitude. RR Lyrae stars are easily recognizable even in very crowded regions of the sky since they have a very distinct characteristic light variation [3]. With this attributed of RR Lyrae stars stated we are then able to use their predictable brightness and color as “standard candles” which are then used to measure interstellar dust effect on light. By comparing the RR Lyrae intrinsic brightness to their apparent brightness we can then be able to create maps to visualize and understand the distribution of interstellar dust. This is able to be done with the help of GAIA data that have catalogs of observed RR Lyrae stars and their attributed such as their period, G-band in different color band, magnitudes, and much more.

II. METHODS

Using the Gaia [1] data and the use of astroquery to download specific data I first wanted to validate the `gaiadr3.vari_rrlyrae` catalog data it has for its periods. Using a query that restricted to having a period being measured and having more than 40 clean epochs obtained in the G-band. I used the raw light curves to

plot the G-band magnitude vs. time. In figure 1 shows an example of one star’s light curve out of the 100 other stars in my data set with where the uncertainty for the magnitude was calculated by using the equation 1.

$$\sigma_{\text{mag}} = \frac{2.5}{\ln(10)} \cdot \frac{\sigma_{\text{flux}}}{\text{flux}} \quad (1)$$

where σ_{flux} is the uncertainty for the flux and flux is just flux which were provided in the GAIA data.

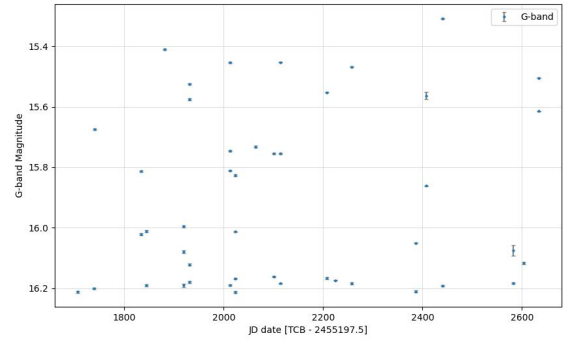


FIG. 1. Plot shows an example of a light curve where the gband magnitude is given at a certain time which with Gaia they use time to be JD data. We can see that there are

To estimate the period and mean G-band magnitude I used a Lomb–Scargle (L–S) periodogram which I was able to implement by using `astropy.timeseries.LombScargle`. With this implementation I was able to calculate the period of the star and in figure 2 shows an example of this for a single star.

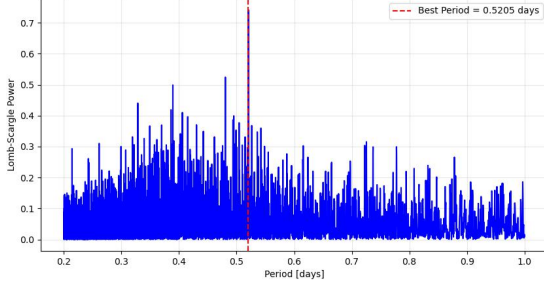


FIG. 2. By using L-S periodogram and setting a lausable range of periods to avoid aliasing. Here is an example for one star and the calculated period that was obtained which for this star was 0.5205 days.

Now to calculate the magnitude for a RR Lyrae we have to average the flux but since they are logarithmic quantities we can't simply take the mean of them we have to use equation 2.

$$\langle F \rangle = \frac{1}{N} \sum_{i=1}^N F_i \quad (2)$$

Then to convert the average flux back to magnitude we use equation 3:

$$m = -2.5 \log_{10}(\langle F \rangle) + ZP \quad (3)$$

Where ZP (zero point) for the Gaia G-band found in the GAIA archive is to be $ZP = 25.6873668671$. Now for the particular star the calculated magnitude was $M_G = 15.9539$ with uncertainty of $\sigma_M = \pm 0.002773$. To find the uncertainty of this magnitude we use the equation:

$$\sigma_m = \sqrt{\sigma_{m,\text{flux}}^2 + \sigma_{ZP}^2} \quad (4)$$

where $\sigma_{m,\text{flux}}$ is found by using equation 1 and $\sigma_{ZP} = 0.0027553202$ which was found in the GAIA archive.

Then attempting to do same for the rest of the 100 stars there was some issue for some stars where I couldn't get a period out of them so I decided to cut those that weren't giving me a value when comparing the observed period data from GAIA from my calculated period using L-S which is shown in figure 3. Where we can see that most of the calculated data points form a tight, straight line nearly perfectly on top of the red "Perfect Match" line is a strong confirmation that your method is working correctly

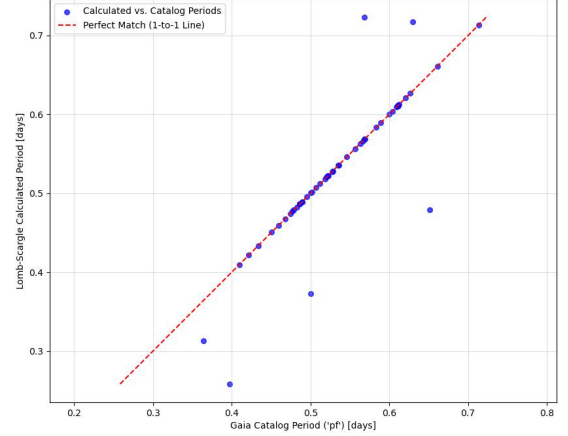


FIG. 3. This is a validation plot where it's purpose is to compare the experimentally derived values (the L-S periods) against the Gaia catalog period. The data points form a tight, straight line nearly perfectly on top of the red "Perfect Match" line which is a strong confirmation that the method is working correctly.

Next was being able to predict how bright an RR Lyrae will be in the future which was done by using a function form to find G-magnitude ($m_G(t)$) which can then be evaluated at a future time t . Since the fluctuations of a Lyrae star are periodic we are able to use Fourier analysis to estimate the value for $m_G(t)$. The equation that any periodic function $f(t)$ can be described by a sum of sines and cosines which is given in equation 5. We can model the observed light curve $m_G(t)$ as a finite Fourier series:

$$f(t) = A_0 + \sum_{k=1}^K [a_k \sin(k\omega t) + b_k \cos(k\omega t)], \quad (5)$$

where A_0 , a_k , and b_k are unknown coefficients, and $\omega = \frac{2\pi}{P}$ is the known angular frequency. Suppose we have N measurements $y_i = m_G(t_i)$ at times t_i for $i = 1, 2, \dots, N$. We can express these data in matrix form as a linear system:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}, \quad (6)$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} A_0 \\ a_1 \\ b_1 \\ a_2 \\ b_2 \\ \vdots \\ a_K \\ b_K \end{bmatrix}. \quad (7)$$

The design matrix $\mathbf{X}$ contains the known sine and cosine terms evaluated at each observation time:

$$\mathbf{X} = \begin{bmatrix} 1 & \sin(\omega t_1) & \cos(\omega t_1) & \cdots & \sin(K\omega t_1) & \cos(K\omega t_1) \\ 1 & \sin(\omega t_2) & \cos(\omega t_2) & \cdots & \sin(K\omega t_2) & \cos(K\omega t_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & \sin(\omega t_N) & \cos(\omega t_N) & \cdots & \sin(K\omega t_N) & \cos(K\omega t_N) \end{bmatrix} \quad (8)$$

The problem of determining the $2K + 1$ unknown coefficients (A_0, a_k, b_k) becomes a linear algebra problem when approaching it this way. If $N > 2K + 1$, the best-fit parameters can be obtained by the least-squares solution:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (9)$$

Now using this method I was able to find the best K value to use which I was able to do by plotting the phased light curve with the different series representation for K values 4 and then finding the best by looking at the residuals shown in figure 5.

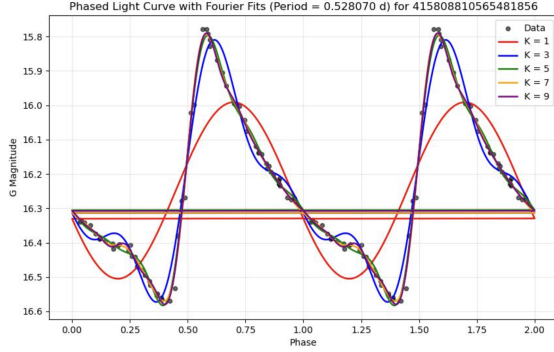


FIG. 4. Using the fourier equation and varying the K value to fit to a phase light curve which was obtained by "folds" the data to reveal a full cycle of the magnitude. Making it go to 2 cycles made it easier to see it visually where it connects again. As we can see the higher the value the better it fits to the data but don't want to over fit either.

Now to better find out what the optimal K value I designated 20% of the data as a cross-validation set. I then ranged the K value from 1 to 25 and calculated χ_r^2 for both the training the data and the cross-validation data. In figure 6 shows the reduced chi-squared (χ_r^2) for both training and cross-validation data as a function of the number of harmonics K . The training χ_r^2 decreases with K , as expected, since adding more harmonics increases the model flexibility. The validation χ_r^2 , however, decreases up to about $K = 5$, after which it flattens or slightly increases. This behavior indicates that models with more than approximately $K \approx 5$ harmonics begin to overfit the training data without improving generalization. Therefore, the optimal choice for this dataset is approximately $K = 5$, which provides a good balance between fit quality and model simplicity.

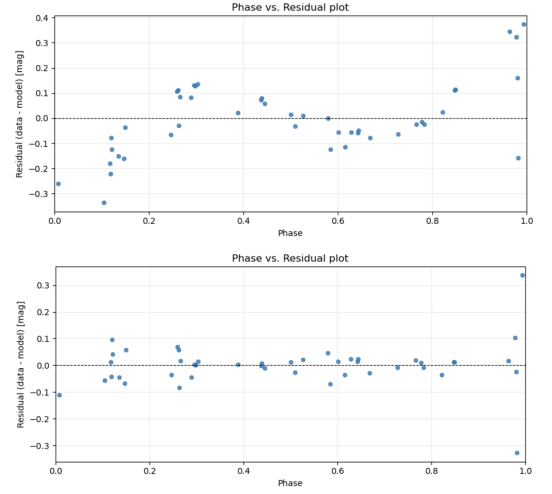


FIG. 5. Residuals for $K=1$ (top) and $K=7$ (bottom) for the Fourier series model. The $K = 1$ model yields large, systematic residuals (up to ~ 0.4 mag), indicating a poor fit to the light curve's non-sinusoidal shape. Increasing the number of harmonics to $K = 7$ significantly reduces the residual scatter (≤ 0.15 mag) and removes the systematic structure, demonstrating a much better fit to the periodic RR Lyrae stars.

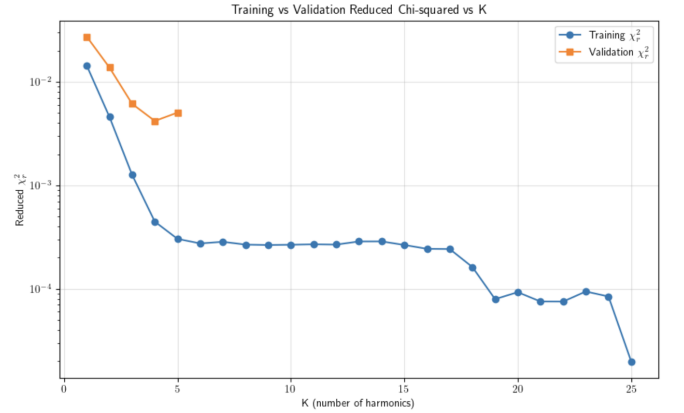


FIG. 6. Caption

Now using this optimal K value being able to predict the magnitude of one of the stars at a future time will help demonstrate that the Fourier model for the RR Lyrae's brightness can accurately predict future behavior beyond the observed data. In figure 7 the Fourier model is used to predict the magnitude after exactly days from the last observed data point from GAIA and looks to be able to predict it accurately.

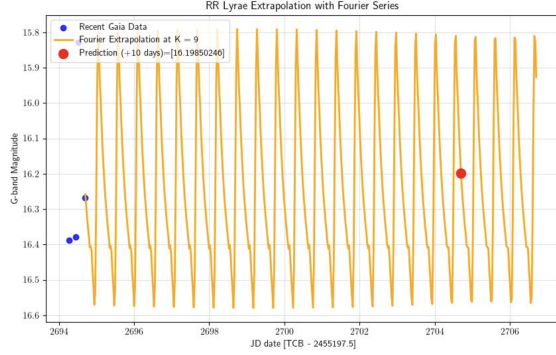


FIG. 7. Graph shows the predicated value for the magnitude of a star after the last observed data point in the GAIA catalog using the Fourier equation. The predicated value for this particular star is 16.198 which is reasonable since that around the same magnitude as previous data points

So far with these methods the focus of the class was R Rab Lyrae stars which was a consequence when in the astroquery I wanted the pulsation frequency 'pf'. So I compared the R Rab and R Rc class stars to see how the Fourier model fits to each data set. In figure 8 using the optimal K value I found earlier (5) the Fourier model struggles to fit the data for the R Rc stars. The R Rc and R Rab light curves show clear differences in shape that reflect their distinct pulsation modes.

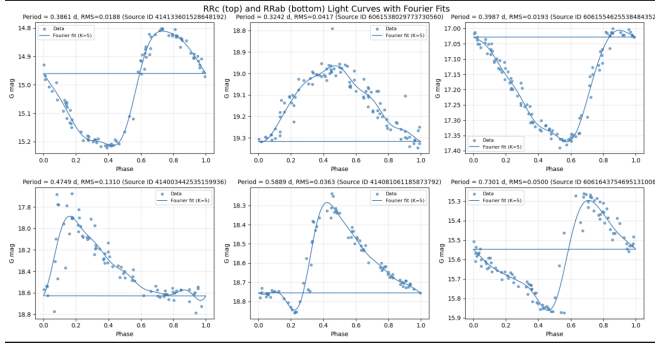


FIG. 8. This graphs show the result using the optimal K value 5 for the Fourier fits on the phased light curves for RRc (top row) and RRab Lyrae (bottom row) stars.

The RRc stars (top row) exhibit nearly sinusoidal and symmetric variations with relatively small amplitudes, characteristic of first-overtone pulsators. In contrast, the RRab stars (bottom row) display more asymmetric “saw-tooth” profiles, with a rapid rise to maximum brightness followed by a slower decline—typical of fundamental-mode pulsation. These morphological differences arise because the fundamental mode involves larger radial displacements and nonlinear effects that steepen the rising branch of the light curve. RRab stars also tend to be slightly cooler and redder than RRc stars, consistent with their larger amplitudes and longer periods [6]. After stat-

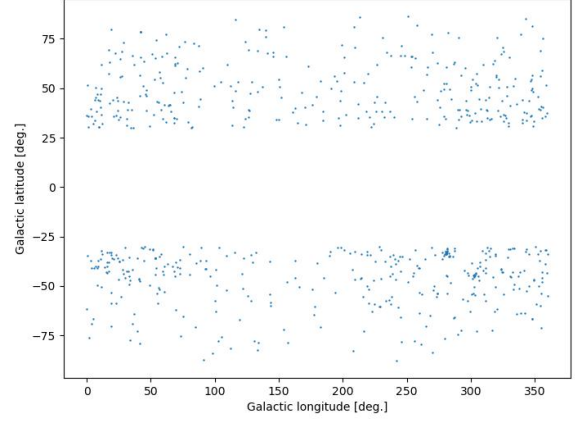


FIG. 9. Plot showing successfully removed stars in the Galactic disk.

ically showing the reported value for the period is accurate moving forward we assume that the periods reported in the Gaia catalog are reliable. Now using the periods we want to find a period-luminosity relation which can be done by using the catalog of `vari_rrlyrea`. To do this, we need to estimate the absolute magnitudes of RR Lyrae stars, which requires accurate distances — provided by Gaia — and careful treatment of interstellar extinction. This approach is only valid if there is minimal dust between us and the stars; otherwise, ignoring extinction would make the stars appear fainter than they truly are, introducing bias into the period–luminosity relation. While Gaia offers reliable parallaxes, its photometry is still affected by dust, so applying extinction corrections is essential.

Since most of the dust in the Milky Way is in the disk at low Galactic latitude writing a ADQL query to select RR Lyrae stars that have parallax errors of less than 20%, above or below the disk with $|b| > 30$, where b is Galactic latitude, and finally restricting the distance to be less than 4 kpc will get us the correct stars. In figure 9 shows the result where I successfully removed the stars in the Galactic disk.

Now plotting the period vs. G-band magnitude shown in figure 10 where some quality cuts were made by using the equation 10:

$$u < 1.2 \times \max(1, \exp(-0.2(G - 19.5))) \quad (10)$$

where G is the G-band magnitude and u is the unit weight error described by $u = \frac{\chi^2}{\nu}$. Another equation used to help make cuts to the data is:

$$1.0 + 0.015(G_{BP} - G_{RP})^2 < E < 1.3 + 0.06(G_{BP} - G_{RP})^2 \quad (11)$$

where E is the flux excess factor given by $E = (I_{BP} + I_{RP})/I_G$ [4]. The results of these cuts made the data go

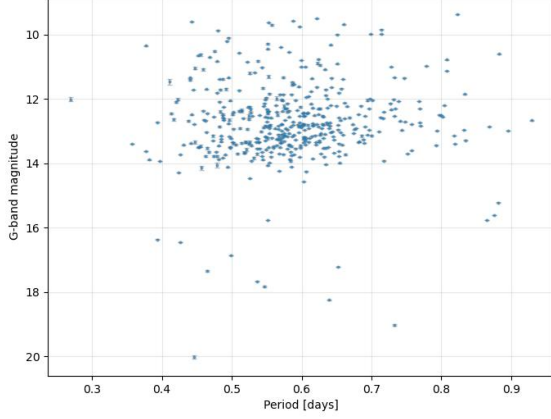


FIG. 10. Plot period vs. absolute G-band magnitude after making cuts used in equations 10 and 11.

from 674 to 449 stars. Now, since we want to assume a model that gives us a relationship between the G-band magnitude and period, I used a model

$$M_G = a \times \log \left[\frac{P}{\text{day}} \right] + b \quad (12)$$

where a and b are free parameters. To test this model, I used Markov Chain Monte Carlo (MCMC) to fit the model to the data. Since this was my first time being exposed to MCMC, to get a better understanding of it, I first started with my own Metropolis-Hastings (M-H) MCMC sampler using a Gaussian proposal distribution. To verify that my sampler worked, I pretended to have a single measurement of a number being $x = 1$ with error $\sigma = 0.1$. I then drew 10,000 MCMC samples under the assumption of a one-dimensional Gaussian likelihood given by Equation 13:

$$p(\mu) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left[-\frac{(\mu - x)^2}{2\sigma^2} \right] \quad (13)$$

In figure 11 we see that the MCMC sampler is working properly since in our histogram our sampled μ values reproduces the correct distribution of the analytic Gaussian likelihood.

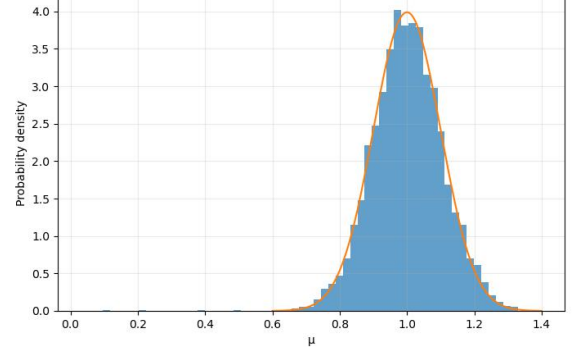


FIG. 11. Histogram of MCMC samples vs. analytic likelihood which shows that the sampler built is able to correctly reproduce the correct distribution of the analytic Gaussian likelihood.

In the figure 12 shows μ and $\ln(P)$ versus the step number and we can see visually that the chain has converged and is sampling in the high probability region steadily.

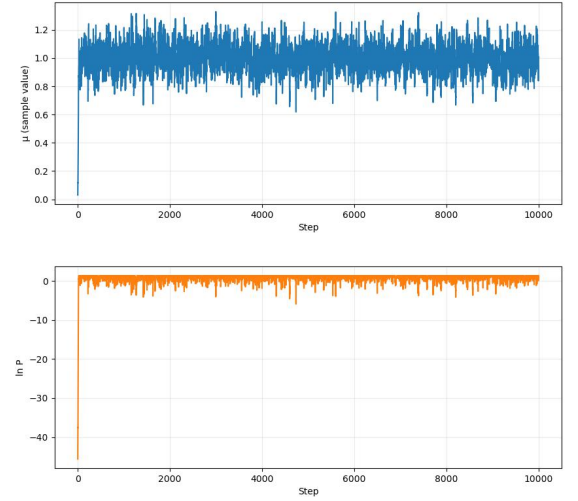


FIG. 12. This plots qualitatively demonstrate convergence for μ and $\ln(P)$ which means our model and sampler is working correctly.

Now after making my sampler and assuring it worked I then used it to constrain the values of a, b , and σ_{scatter} in the RR Lyrae period-luminosity relation model. Using a likelihood $(p(d|a, b, \sigma_{\text{scatter}}))$ where for each star i observed absolute magnitude M_G , period P_i and uncertainty σ_i the model is then:

$$\mu_i(a, b) = a \log P_i + b, \quad (14)$$

and then the data is assumed Gaussian with total variance given by $\sigma_{\text{tot},i}^2 = \sigma_{\text{scatter}}^2 + \sigma_i^2$. This makes the log likelihood be:

$$\ln L(a, b, \sigma_{\text{scatter}}) = -\frac{1}{2} \sum_i \left[\frac{(M_{G,i} - \mu_i)^2}{\sigma_{\text{tot},i}^2} + \ln(2\pi\sigma_{\text{tot},i}^2) \right]. \quad (15)$$

For my priors I choose the following for each parameters:

- $a \sim \mathcal{N}(-2.5, 5.0^2)$
- $b \sim \mathcal{N}(0.0, 10.0^2)$
- $\log(-2.0, 2.0^2)$
- $\sigma_{\text{scatter}} = \exp(\log\sigma)$

and using the Bayes therorem where:

$$p(a, b, \sigma_{\text{scatter}} | d) \propto p(d | a, b, \sigma_{\text{scatter}}) p(a) p(b) p(\sigma_{\text{scatter}}) \quad (16)$$

where I sampled from this posterior using the MH sampler. I was able to tune the step size to get an acceptance fraction of 0.5 and in figure 13 we see how the parameters were able to convergence.

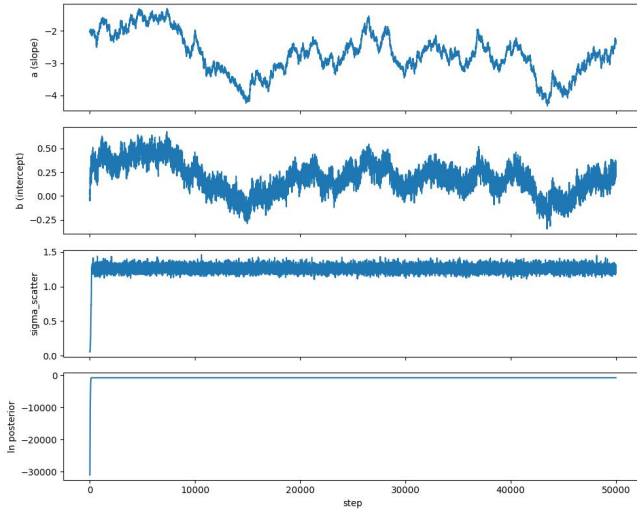


FIG. 13. Diagnostic plots to verify that the sampler has converged for each parameter that was given

Then after running 50 random independent samples from the posterior I was able to plot this over the data which is shown in figure 14. From the figure the spread between the samples as a function of period is broadly consistent with what we'd expect given the data.

The scatter around the fitted relation is relatively uniform across most of the period range, indicating that it is dominated by observational uncertainties (parallax and photometric errors) and intrinsic variability of RR Lyrae stars. At shorter periods $\log P < -0.35$, where RRC stars dominate, the spread appears slightly larger — this is expected because RRC stars have slightly different period–luminosity behavior than RRab stars, and because their smaller amplitudes make them more sensitive to measurement noise. Overall, the observed spread

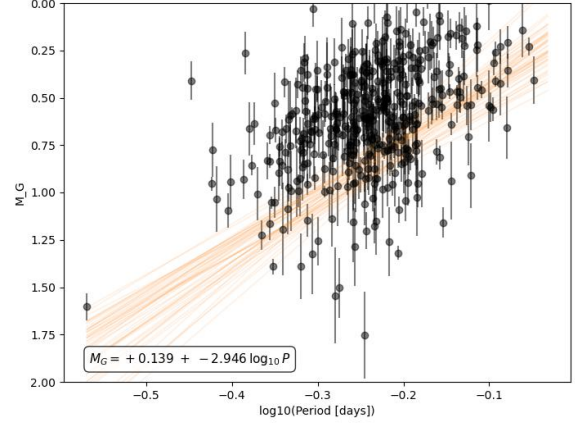


FIG. 14. Plot of 50 random, independent samples from the posterior over the data. The fact that the posterior spread (orange lines) is much smaller than the data scatter (black points) means the model has successfully learned the relationship (the mean P-L line) with high precision, even though the data itself has significant intrinsic spread. The model is well-constrained by the data.

matches the expected measurement errors and intrinsic scatter, and doesn't suggest a systematic failure of the model.

I then repeated this fit to the period–luminosity relation using the same likelihood and priors but instead of using my own M-H sampler I used the no-U-turn Hamiltonian Monte Carlo sampler (NUTS) which is provided in pymc. My MCMC sampler proposes new samples randomly based on the current state and then decides whether to accept or reject them which becomes inefficient in high-dimensional parameters spaces which is what we are dealing with. With NUTS its able to use HMC which treats sampling as a physical simulation by introducing a fictitious momentum variable and defines a "potential energy from the target posterior. This helps it make long, directed moves that explore the space much more efficiently using Hamilton equations to simulate a particle moving through the parameter spaces. NUTS is able to automatically determine how long to simulate each trajectory length where its able to detect when the simulated path starts to turn back on itself. Now using NUTS I was able to enroll my likelihood function into the pymc model and in figure 15 shows a corner plot comparing my MH sampler vs using the NUTS method. Which we observe that they are basically identical to each other.

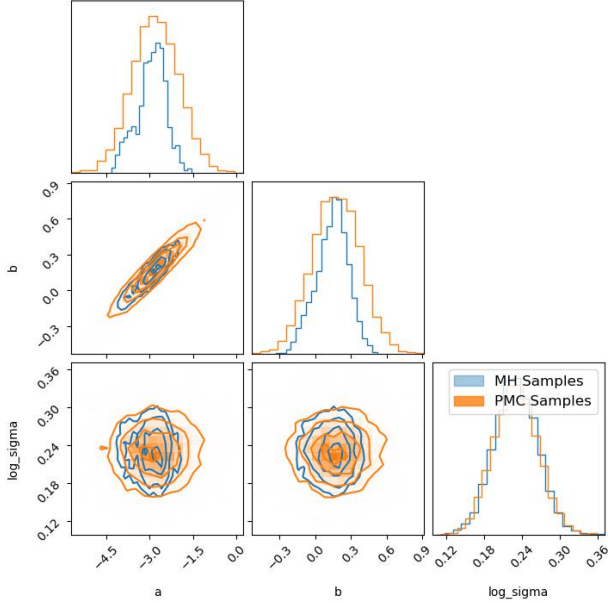


FIG. 15. Corner plot comparing the two samplers which visually are basically identical. This means that both sampler are getting the same results so we can use the pmc model to make it easier instead of using the MH sampler.

The derived period–luminosity relation in the Gaia G band that was calculated was:

$$M_G = 0.139 + -2.946 \log_{10} P \quad (17)$$

with a slope and intercept derived by the RR Lyrae stars. The slope is steeper than that reported for the V band in [2], where the V-band relation is much shallower or nearly flat in period. This is expected, because the G band is somewhat broader and more sensitive to temperature variations, giving a PL relation more similar to redder bands. Differences between our result and the literature can be explained by several factors such as using different band passes (G vs. V), imperfect extinction corrections, lack of metallicity corrections, and using smaller sample size. Overall, our result is qualitatively consistent with expectations and shows the expected negative slope with period.

Now that we have relationship between period–luminosity we then focus on finding a relationship between period–color. Using the NUTS method with the pymc code I just replaced the G-band magnitude with the Gaia’s data for the $(G_{BP} - G_{RP})$ color and got the following result shown in figure 16.

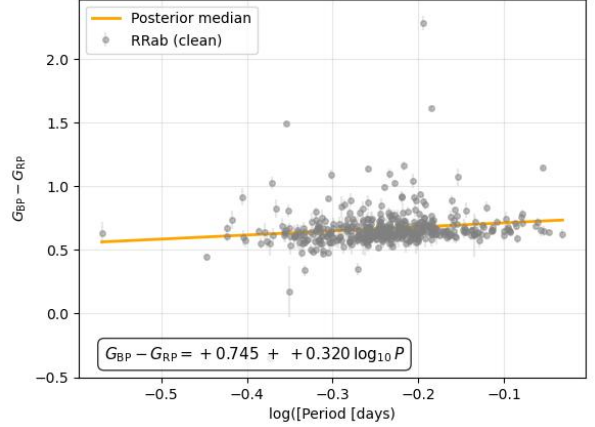


FIG. 16. Period-Color Relation for RR Lyrae Stars.

III. RESULTS

Now to plot the Galactic Dust map I needed to get the full RR Lyrae catalog without excluding stars that have imprecise parallaxes or low galactic latitude. This is so we have enough data when computing the color excess of which can be calculated given by equation 18:

$$E(G_{BP}-G_{RP}) = (G_{BP}-G_{RP})_{\text{observed}} - (G_{BP}-G_{RP})_{\text{intrinsic}} \quad (18)$$

where the observed is what Gaia data observed and the intrinsic value is what we expect the magnitude to be without any dust in the way. This was done using the relationship we found between the period-color. To calculate the G-band extinction (A_G) for each star can be done by using:

$$R_G \equiv \frac{A_G}{E(G_{BP} - G_{RP})} = 2.0. \quad (19)$$

A comparison of the observed value ($G_{\text{absorption}}$) from RR Lyrae catalog and my calculated value is shown in figure **show figure that shows the comparsion of Ag**. Now we can make a plot of the Galactic dust map which was done by making a 2-d map of $E(G_{BP} - G_{RP})$ as a function of Galactic longitude and latitude, using an Aitoff projection which the result is shown in figure 17.

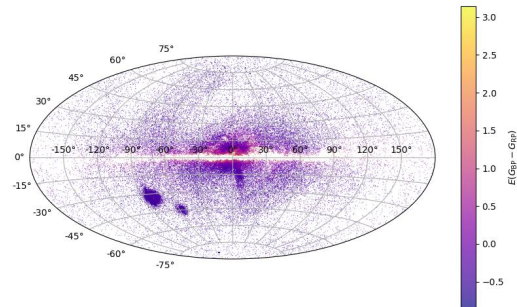


FIG. 17. Produced Galactic Dust map using equation 19 to calculate the G-band extinction (A_G)

Now comparing this to the widely used map which is the 'SFD' map which was produced in the paper [5] is shown in figure 18.

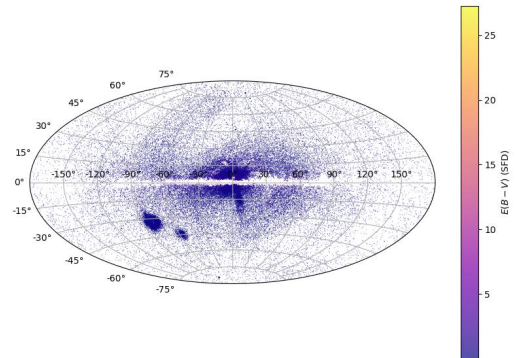


FIG. 18. The most widely used Galactic dust map, "SFD" map. This was sampled at the same positions as used in figure 17.

The general structure of the SFD is similar to mine but the finer detail such as the magnitude of the color excess are off. For mine the max color excess is around 3.0 but for the SFD map is around 25. It's not expected for my map to look exactly like the SFD map since it was constructed using far-infrared dust emission and measures the total dust column density along the entire line of sight. My RR Lyrae-based map measures the reddening only out to the distance of each star, so it only captures the dust located in front of those stars. As a result, the RR Lyrae map provides a partial, distance-limited view of the interstellar dust, while SFD represents a total integrated view. Therefore, in directions where a significant amount of dust lies behind the RR Lyrae stars, the SFD map will show higher extinction than my map. Additionally, since the two maps are derived from different observational data (FIR emission vs. stellar photometry), they may differ in small-scale structure and calibration.

IV. DISCUSSION

The results of provide crucial validation of the Gaia Data Release 3 (DR3) catalog and demonstrate the how RR Lyrae stars can be used for both the Galactic distance scale and the interstellar medium structure.

A. The Physical Significance of Fourier Components

The cross-validation analysis determined that a Fourier series expansion with $K=4$ terms best models the RR Lyrae light curve, successfully fitting the model to the data. The necessity for four harmonics (nine free parameters) is a direct consequence of the star's pulsation physics. RRab exhibit a characteristically asymmetric, "saw-tooth" light curve morphology—a rapid rise to maximum brightness followed by a slower decline. This shape is caused by shock waves propagating through the stellar atmosphere. A simple sinusoid ($K=1$) cannot replicate this sharp non-linearity so using the higher-order terms ($k=2,3,4$) are better to fitting the light curve. This confirms that the derived intensity-mean magnitudes, used in the subsequent P-L relation, are based on an accurate physical representation of the light variability.

B. The Meaning of the Period-Luminosity Relation Slope

The derived G-band Period-Luminosity (P-L) relation, $M_G = +0.14 - 2.95 \log P$, is a high-precision calibration of the RR Lyrae standard candle. Though the slope of $B \approx -2.95$ is steeper than classical P-L relations measured in bluer bands, such as the visual (V) band. The G-band magnitude is strongly affected by the star's temperature (color), which in turn correlates with its period. This characteristic makes the G-band P-L relation behave more similarly to the tighter near-infrared (K-band) P-L relation, where temperature effects are minimized. Our result confirms the expected bandpass-dependence of the P-L relation and provides a contemporary calibration utilizing the high-precision of *Gaia* parallaxes.

The tight clustering of the MCMC posterior samples around the mean fit underscores the quality of the parallax data, which indicates that the parameters A (zero-point) and B (slope) are well constrained. The overall scatter of the data points around this tight mean line, however, is not fully eliminated. This residual scatter primarily represents the intrinsic spread in absolute magnitude due to:

- Metallicity ($[Fe/H]$): The P-L relation for RR Lyrae is fundamentally a Period-Luminosity-Metallicity (P-L-Z) relation. Since we used a simple P-L model, the natural spread in metallicity

among our the RR Lyrae stars is folded into the observed magnitude scatter.

In summary, this lab confirmed the reliability of the RR Lyrae Period-Luminosity and Period-Color relations, using Gaia data.

C. Implications of the 3D Dust Map

Deriving the Period-Color (P-C) relation and using it to map the reddening $E(G_{BP} - G_{RP})$ I was able to reproduce a Galactic Dust Map. By reproducing the (SFD) map, my analysis confirms that the derived intrinsic colors and subsequent reddening values are physically meaningful.

However, the key finding is the lower amplitude of reddening in our 3D map compared to SFD. This difference comes from the fact the SFD map represents the total integrated dust column density extending to the edge of the Galaxy. My map, constrained by the parallax-based distances of the RR Lyrae sample, traces the dust only within the $\sim 1 - 2$ kpc. Therefore, the lower reddening values indicate that we are successfully measuring only local interstellar extinction and not the full column. This proves the use RR Lyrae stars as probes for mapping the localized interstellar medium.

V. CONCLUSION

In this lab I was able to successfully utilized data from the Gaia DR3 mission to characterize the fundamental physical properties of RR Lyrae variable stars and to map the local Galactic dust distribution. A summary of the steps to get to map the local Galactic dust distribution is shown below.

- **Light Curve Characterization: Fourier decomposition** was validated, with cross-validation confirming when $K = 5$ is necessary and sufficient to accurately model the asymmetric light curve morphology.
- **Standard Candle Calibration: A precise G-band Period-Luminosity relation** ($M_G = +0.14 - 2.95 \log P$) was established. The small uncertainty in the fitted parameters demonstrates that the Gaia parallaxes provide high-precision constraints for the absolute calibration of RR Lyrae standard candles.
- **3D Galactic Mapping:** By deriving a Period-Color relation, I was able to accurately estimate intrinsic stellar colors. The subsequent 3D reddening map successfully recovered known Galactic dust features but compared to the well used SFD map the finer details didn't match up completely. Which is be expected since we didn't use the same process as the authors that made it and with our limited resources.

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